

$$1: \frac{x^2-16}{x+4} \text{ 为0时 } x = \underline{\hspace{2cm}}$$

$$2: \frac{2x}{1x+\frac{1}{2}} \text{ 有意义, 则 } \underline{\hspace{2cm}}$$

$$3: \frac{x}{2} = \frac{y}{4} = \frac{z}{6} \neq 0, \text{ 求 } \frac{2x-y+z}{x-2y-z}$$

$$4: \langle 1 \rangle \frac{m^2-4}{m^2-4m+4} \div \frac{2}{2-m}$$

$$\langle 2 \rangle \frac{y^2}{x-y} \cdot \frac{2x-y}{x} \div \frac{4y}{x}$$

$$\langle 3 \rangle \frac{4x+3}{x^2-3x} + \frac{15}{3x-x^2}$$

$$\langle 4 \rangle \frac{4x^2-4xy+y^2}{2x-y} \div (4x^2-y^2) \cdot \frac{1}{2x+y}$$

$$\langle 5 \rangle \left(\frac{a-b}{ab}\right)^2 \div \left(-\frac{b-a}{a}\right)^3 \cdot (a-b)$$

$$\langle 6 \rangle \frac{2}{y^2-1} - \frac{2}{y+1} - \frac{1}{y-1}$$

$$\langle 7 \rangle \frac{x-2}{x+2} - \frac{x-2}{x^2+2x} + 2$$

$$\langle 8 \rangle \left(1 - \frac{1}{x+2}\right) \div \frac{x+1}{x^2+4x+4}$$

$$\langle 9 \rangle \frac{a+2b}{a+b} - \frac{a-b}{a-2b} \div \frac{a^2-b^2}{a^2-4ab+4b^2}$$

$$\langle 10 \rangle \left(\frac{a+1}{2a-2} - \frac{1}{2a^2-2}\right) \div \frac{a}{a^2-1}$$

$$\langle 11 \rangle \frac{x^2-2x+1}{x^2-1} \div \left(1 - \frac{1}{x+1}\right)$$

$$\langle 12 \rangle \left(\frac{a^2}{a+1} - a + 1\right) \div \frac{a^2-1}{a^2+2a+1}$$

$$\langle 13 \rangle \frac{1}{(x-1)(x-2)} + \frac{1}{(x-2)(x-3)} + \dots + \frac{1}{(x-2022)(x-2023)}$$

$$\langle 14 \rangle A = \frac{x}{x^2-y^2} \quad B = \frac{y}{y^2-x^2}$$

$$\langle 1 \rangle \text{ 求 } A+B \quad \langle 2 \rangle \text{ 求 } A-B$$

$$\langle 3 \rangle \text{ 若 } A+B=2, A-B=-1, \text{ 求 } x, y$$

$$\langle 15 \rangle \text{ 若 } a + \frac{1}{a} = 2, \text{ 求 } \langle 1 \rangle a^2 + \frac{1}{a^2} \quad \langle 2 \rangle a^4 + \frac{1}{a^4}$$

$$16: \frac{|a|}{a-a^2} = \frac{1}{a-1} \text{ 求 } a \text{ 范围}$$